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Marangoni effect in laser deep penetration welding of steel

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In welding, the resulting weld-seam geometry may vary significantly although using constant process parameters and steels with the same material number. One likely reason for this are small variations in the concentration of sulfur, phosphorus, oxygen, and other chemical elements that are well within the tolerance of the standard of a specific alloy. These substances act as surfactants and even marginal changes strongly effect the temperature-dependent coefficient of surface tension. In simulations of conventional electric arc welding and laser heat conduction welding, the effect of the temperature-dependent coefficient of surface tension (Marangoni effect) has been identified as one of the primary driving forces of the liquid melt. In laser deep penetration welding simulations this effect has been widely neglected, so far. In this contribution, simulations of flow fields in the weld pool resulting from different temperature dependencies of the coefficient of surface tension are presented. The simulations show the same effect in laser deep penetration welding that is already known in gas tungsten arc welding. Consequently, it becomes clear that the chemical specifications for many steel alloys are too crude for reliable predictions of the seam geometry in laser deep penetration welding. © *2001 Laser Institute of America.*

Key words: laser welding, Marangoni effect, thermocapillary convection, weld-pool flow, heat transfer, surface tension, melt-flow simulation

I. INTRODUCTION

In laser welding of steel, there are many parameters that influence the weld-pool geometry and ultimately the shape of the weld seam. Most deciding parameters are welding speed, laser power, focus diameter, focus position with respect to the workpiece, beam quality, beam polarization, material properties and geometry of the workpiece, filler wire, condition of the metal surface, and shielding gas. The welding process results from a highly complex interaction of different physical phenomena determined by these parameters (see Fig. 1). Consequently, the understanding of the welding process in its entireness is an extremely difficult task. But in order to choose the ''right'' parameters for a prescribed seam shape, it is important to understand its underlying physical effects. One possible procedure to get to applicable results is called ''synthetic modeling'' (see Ref. 1). This means concentrating on just one or a few parameters and making simplifications for the remaining ones. In simulations, one can identify mechanisms which cannot be isolated and separately studied in experiments. Examples for this investigation method applied to processes involved in laser welding are the absorption of polarized laser light within the keyhole, the optical properties of a plasma plume, and the influence of welding speed on the flow field around the keyhole, which can be found in Ref. 2. A further example is the transient keyhole behavior described in Refs. 3–5.

This contribution concentrates on the Marangoni effect, which is caused by a spatially varying coefficient of surface tension. In laser deep penetration welding, this effect has not been paid much attention to although it is one of the dominating driving forces of the melt in laser heat conduction welding[6,7] and gas tungsten arc welding.[8] The simulations presented here demonstrate a comparable importance in laser deep penetration welding.

II. INTRODUCTION TO THE MARANGONI EFFECT

A. Experimental observations

Investigations of the weld-seam geometry by systematic variations of the sulfur content of steel are rarely found in the literature. The effect of sulfur and other surface-active elements on the weld-seam geometry is widely recognized for gas tungsten arc welding (GTAW). In a recent publication of Pierce, Burghardt, and Olson,[9] the effect of sulfur content in type 304 stainless steel has been demonstrated experimentally for GTAW and electron-beam welding (EBW) (see Fig. 2). Even though GTAW and EBW are quite different types of welding processes, the sulfur content plays an important—and essentially the same—role in both of them. In GTAW,

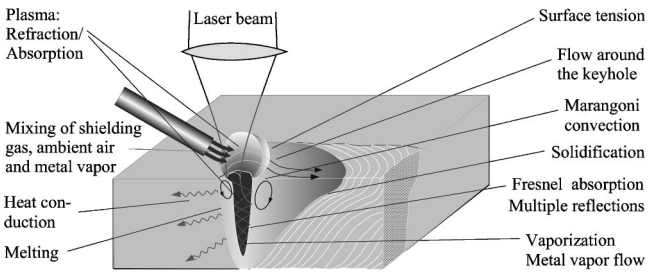


FIG. 1. Interaction of different physical phenomena in laser welding.

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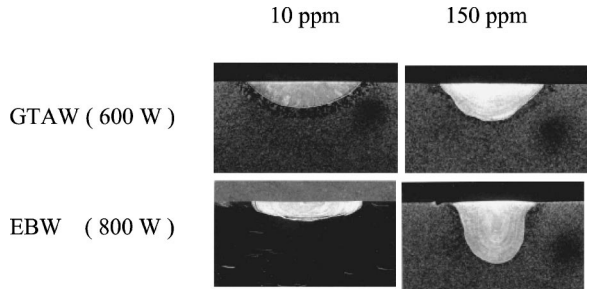


FIG. 2. Weld-seam geometries in gas tungsten arc welding (GTAW) and electron-beam welding (EBW) of steel at different sulfur contents, see Ref. 9. All other parameters were kept constant.

the penetration depth roughly doubles by slightly increasing the sulfur content from 10 to 150 ppm. In EBW, the penetration depth even triples with the same increase in sulfur. Since the EBW process and the laser deep penetration welding (LDPW) process are closely related, one can expect a considerable effect here as well.

Parallel to our numerical investigations, Aidun and Martin[10] investigated the effect of the sulfur content in type 316 stainless steel on welding results in laser deep penetration welding (see Fig. 3). The base material on the left in Fig. 3 has a sulfur content of 10 ppm and on the right of 60 ppm. Even though the differences in sulfur content are much smaller than in Pierce's publication, the experimental results confirm the relation between sulfur content and penetration depth. To get some insight about the flow structure of this LDPW process, we have chosen numerical simulations as an investigation method. Both the model and the results are presented in Sec. III.

B. Marangoni effect in spot welding

Before the complex flow field in laser deep penetration welding will be presented in Sec. III, the Marangoni effect is explained for conduction mode spot welding, which exhibits a much simpler flow field with radial symmetry.

In the left example of Fig. 4, a radially increasing coefficient of surface tension $\gamma(r)$ has been considered. Therefore, the surface flow moves towards the outer boundaries of the weld pool. For demonstration, a small volume element close to the surface is enlarged. The horizontal line atop the volume element represents the free surface dA above the bulk fluid. On the right side of the volume element, the coefficient of surface tension is larger than on the left. Thus, the surface is accelerated towards the outer boundaries. The bulk fluid just underneath the surface is accelerated by friction in

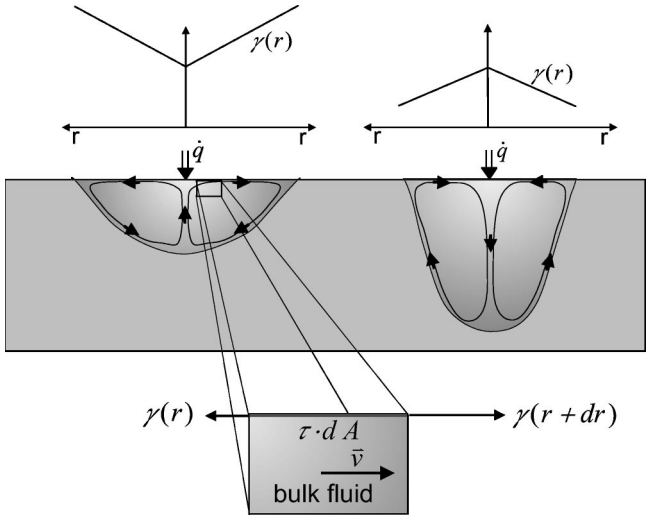


FIG. 4. Weld-pool flow and convective heat transport in spot welding (schematically).

the same direction. In the case of a heat source $\dot{q}$ in the middle of the weld pool, convection transports heat to the sides of the weld pool. Due to this convective heat transport towards the outer pool boundary heat conduction into the solid is increased and the weld-pool diameter increases.

In contrast to this situation, a radially decreasing coefficient of surface tension has been considered in Fig. 4 on the right. In this case, the surface flow moves in the opposite direction towards the middle of the pool. If there is a heat source $\dot{q}$, heat is convected to the weld-pool bottom along the center line increasing the temperature and the melting rate at the weld-pool bottom. As a result a deeper and narrower pool geometry than in the case of a radially increasing coefficient of surface tension is formed.

C. Gradient of coefficient of surface tension

The Marangoni effect depends on the gradient of the coefficient of surface tension and induces shear stress τ_s at a free surface. In turn, the coefficient of surface tension depends on the temperature and on the concentration of additives in the melt. For steel, elements such as sulfur, phosphorus, oxygen, selenium, and telurium are known to be surface active.[11] Others, like aluminum and calcium, are known to chemically combine with surface-active agents and consequently neutralize their effect.[10] Both types of chemical elements can change the spatial distribution of the coefficient of surface tension γ and, therefore, can change the gradient within the surface $\nabla_s \gamma$ as well. The exact differential for the

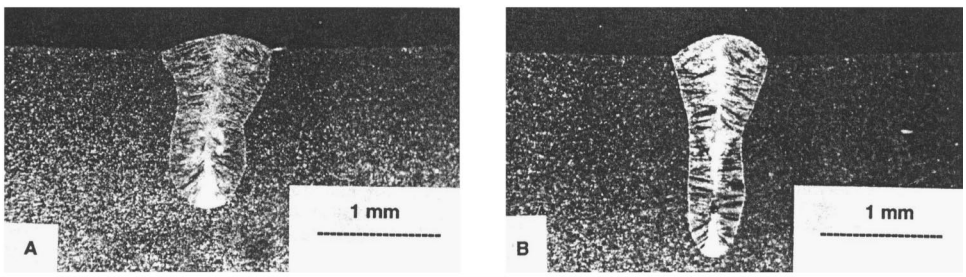


FIG. 3. Laser beam bead-on-plate welds for (a) steel 316 (10 ppm sulfur contents) and (b) steel 316 (60 ppm sulfur contents) using argon as a shielding gas according to Aidun and Martin (Ref. 10).

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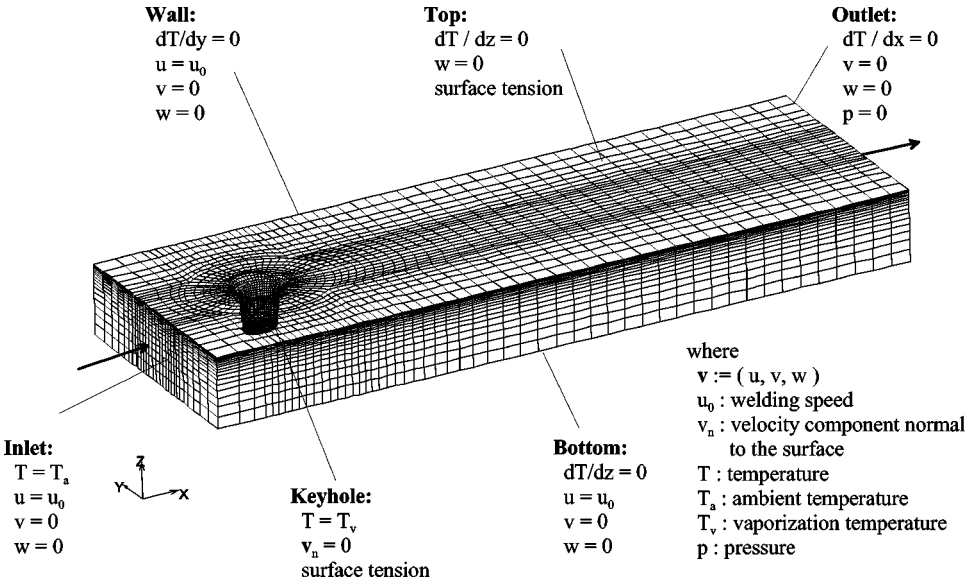


FIG. 5. Mesh with boundary conditions (90 000 elements).

coefficient of surface tension γ at the free surface leads to

$$\nabla_s \gamma(T,c_i) = \tau_s = \underbrace{\left.\frac{\partial \gamma}{\partial T}\right|_{c_i}}_{1} \cdot \underbrace{\nabla_s T}_{2} + \sum_i \underbrace{\left.\frac{\partial \gamma}{\partial c_i}\right|_{T,c_j(j\neq i)}}_{3} \cdot \underbrace{\nabla_s c_i}_{4}, \tag{1}$$

where T is the temperature and c_i the concentration of the ith surfactant species. The derivatives labeled ''1'' and ''3'' are material properties, whereas terms ''2'' and ''4'' are process dependent. They can be influenced, e.g., by filler wire and shielding gas.

Unfortunately, experimentally obtained material properties for $\gamma(T)$ are rare and mostly exist for pure metals only. Yet, there are models that treat the coefficient of surface tension of alloys theoretically. As an example, a model based on a combination of Gibbs and Langmuir absorption isotherms was used by DebRoy[12] for numerical simulations of conduction mode laser welding. Our simulations are based on a very similar model of Vitol and Orlova.[13]

III. NUMERICAL SIMULATION OF THE MELT FLOW

A. Numerical model

After discussing the basics of the Marangoni effect in the last section, the laser deep penetration welding process will be presented in the subsequent sections.

For welding simulations, the correct transport of heat is a decisive factor. Possible mechanisms are radiation, heat conduction, and heat convection. Compared to the high intensity laser radiation the thermal radiation of the weld pool does not seem to be very important. Heat conduction and heat convection play a much larger role. There are many publications which treat the heat conduction within the workpiece, numerically or analytically, e.g. Ref. 14. Although it is generally recognized that the weld-pool flow may reach speeds of several meters per second, which may lead to considerable convective heat fluxes, there are hardly any simulations in laser deep penetration welding that try to account for it. Most authors concentrate on the keyhole dynamics and treat the flow around the keyhole in a very general way without solving the Navier–Stokes equations.[4] Usually, the weld-pool flow behind the keyhole is not treated at all.

The simulations presented in this article were conducted because the Marangoni effect has not been investigated in combination with a steam capillary, which does not exist in conventional welding. This capillary (also referred to as the ''keyhole'') is caused by high vaporization rates of the base metal as the high intensity laser beam hits the surface of the workpiece. When the laser beam advances, the molten base material in front of the keyhole has to flow around the keyhole and, therefore, the velocity of the melt passing the keyhole is faster than the welding speed. This flow may interfere with the Marangoni induced flow in the weld pool behind.

To solve this problem numerically, a computational domain has to be defined and split into small volumes. At the boundaries of this computational domain, boundary conditions have to be applied which represent the effect of the process within the computational domain on its vicinity and vice versa. Such a numerical grid including boundary conditions is displayed in Fig. 5. The laser beam is fixed in the frame of reference with the workpiece moving from the entity ''inlet'' towards the entity ''outlet.'' In this paragraph the assumptions and simplifications of our model are listed which have been used for the subsequent simulations:

- stationary Navier–Stokes equations as governing flow equations for the liquid and solid phase;
- fixed keyhole geometry (no inclination) and fixed weld-pool surface (the keyhole geometry at the surface was observed in experiments at a welding speed of 2 m/min and 3 kW of laser power);
- temperature dependence of the coefficient of surface tension γ;
- homogeneous surfactant concentration throughout the melt, therefore, terms 3 and 4 of Eq. (1) are not included;
- phase change is modeled by temperature-dependent properties (weld-pool geometry is part of the simulation result);

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TABLE I. Geometric, thermodynamic, and technological input data for the simulations.

Welding speed: $u_0 = 2$ m/min.
Keyhole properties: major semiaxis at surface $a = 0.475$ mm, minor semiaxis at surface $b = 0.363$ mm, depth $t_{kh} = 4.30$ mm, inclination $\alpha = 0°$, and cone angle $\beta = 6°$.
Characteristic temperatures: ambient temperature $T_0 = 293$ K, solidus temperature $T_s = 1825$ K, liquidus temperature $T_l = 1849$ K, and vaporization temperature $T_v = 2773$ K.
Workpiece: material: mild steel, sheet thickness $t = 6$ mm, density $\rho = 7800$ kg/m^3, coefficient of heat conduction $\lambda = 41.0$ kg m/(K s), viscosity of solid $\mu_s = 1.0 \times 10^7$ kg/(m s), and viscosity of liquid $\mu_l = 4.3 \times 10^{-3}$ kg/(m s). Coefficient of specific heat of solid $c_{p_s} = 800$ J/(kg K), of solidus–liquidus region $c_{p_{s-l}} = 1236$ J/(kg K), and of liquid $c_{p_l} = 700$ J/(kg K). Slope of coefficient of surface tension: negative $\partial\gamma/\partial T = -5.84 \times 10^{-4}$ N/(m K), and positive $\partial\gamma/\partial T = 5.84 \times 10^{-4}$ N/(m K).

- vaporization temperature at the keyhole wall, although temperature might be lower at the rear keyhole wall and higher at the front keyhole wall;
- constant density ρ (no buoyancy), coefficient of heat conduction λ and vaporization temperature T_v;
- no mass flow through keyhole wall due to vaporization; and
- no friction at the keyhole wall.

Due to the assumptions of stationarity and the fixed surfaces, transient melting instabilities at the front keyhole wall as mentioned by Matsunawa and Semak[4] are suppressed. These transient instabilities can lead to locally high ablation pressures but the amplitude of these instabilities decreases while their frequency increases with increasing welding speed. At moderate and high welding velocities a quite stable front keyhole wall is attained for steel.

In several models,[4,5] the ablation pressure and other pressures are calculated and then prescribed on a temporary weld-pool surface. Our model takes the opposite approach: The (stationary) keyhole geometry is given and the pressure distribution to keep this keyhole in its shape and position is part of the solution.

Since the shape of the keyhole and the free surface are given and kept fixed, the absolute values of the coefficient of surface tension are unimportant. In order to investigate the impact of the Marangoni effect on the flow field, a simulation was conducted at a negative $\partial\gamma/\partial T$ which was kept constant according to Table I. To investigate the effect of a different $\partial\gamma/\partial T$ on the melt-flow field, calculations were performed with a positive $\partial\gamma/\partial T$ as well by just switching signs. Both positive and negative $\partial\gamma/\partial T$ are within an experimentally verified range for different steel alloys.

If, in addition to the temperature-dependent terms 1 and 2 of Eq. (1), the concentration-dependent terms 3 and 4 were included in the model, too, the shielding gas flow would have to be simulated as well, since it could change the concentration of surfactants at the weld-pool surface. The resulting surface tension gradients can lead to an excitation of surface waves which again interact with the shielding gas. In the literature this is referred to as interfacial turbulence.[15,16]

The technological, geometric, and thermodynamic properties for the simulations are listed in Table I.

B. Simulation results

1. Negative $\partial\gamma/\partial T$

Results of the numerical simulation of the LDPW process for the case of a negative $\partial\gamma/\partial T$ are presented in Figs. 6 and 7. This case of a negative $\partial\gamma/\partial T$ resembles the coefficient of surface tension of pure iron. The upper part of Fig. 6 shows the simulated isotherms and the lower part the local velocity vectors at the surface of the weld pool. Due to the temperature dependence of $\gamma(T)$, the coefficient of surface tension is smaller at the keyhole than it is at the outer weld-pool boundary as described for spot welding in Sec. II B. As a consequence, the surface tension accelerates a thin fluid layer from the keyhole region to the outer weld-pool boundary. The maximum velocity at the surface occurs close behind the keyhole. This is also the highest velocity in the entire weld pool and reaches 114 m/min, which is 57 times the welding speed. Looking at the isotherms in Fig. 6, it is striking that the radial spacing between the isotherms is almost constant on most of the weld-pool surface. This means that the temperature gradient in the radial direction is almost constant as well. The reason for this will become more ob-

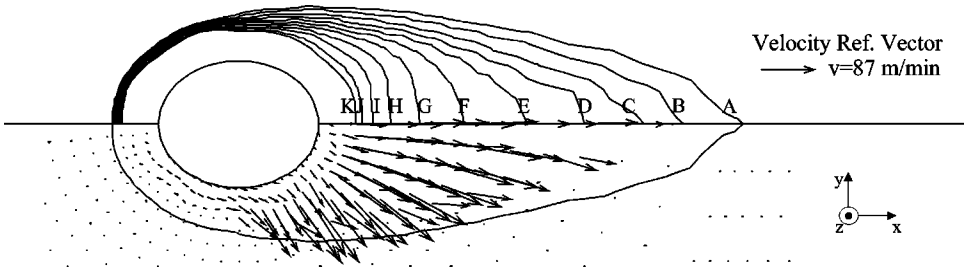


FIG. 6. Temperature distribution (represented by isotherms) and flow field at the surface in case of negative $\partial\gamma/\partial T$. Temperatures of isotherms: $T_A = 1825$ K, $T_B = 1900$ K, $T_C = 2000$ K, $T_D = 2100$ K, $T_E = 2200$ K, $T_F = 2300$ K, $T_G = 2400$ K, $T_H = 2500$ K, $T_I = 2600$ K, $T_J = 2700$ K, and $T_K = 2770$ K (process and material parameters according to Table I).

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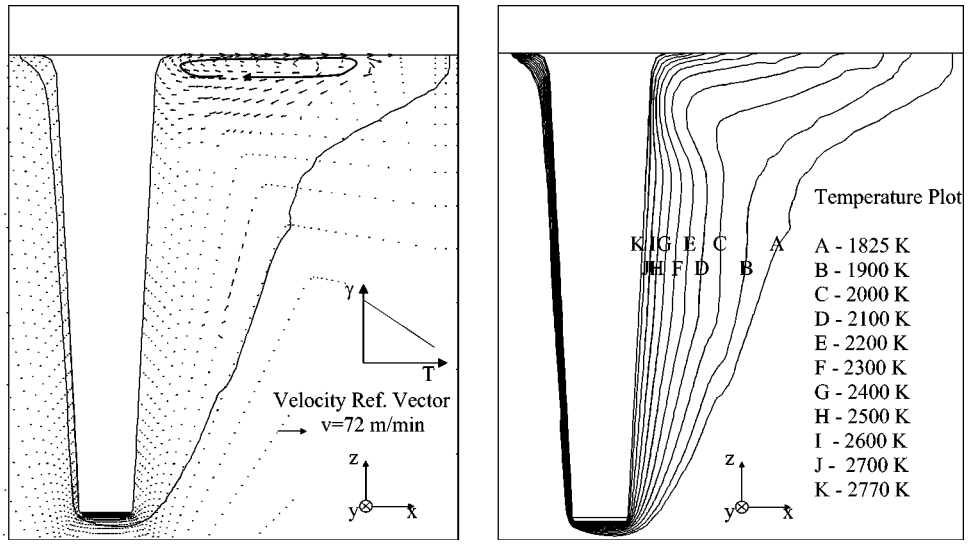


FIG. 7. Velocity vector plot and solidus isotherm in longitudinal section (left) and corresponding isotherm plot (right) in the case of negative $\partial\gamma/\partial T$ (corresponding to Fig. 6).

vious by looking at the flow field in longitudinal section in the plane of symmetry in Fig. 7. On the left, the velocity vectors are shown, whereas on the right one can see the temperature field which is represented by isotherms.

In the case of a simulation without convection, the isotherms would be very closely spaced near the keyhole (large temperature gradient) with an approximately exponentially decreasing temperature gradient towards the outer boundary. Therefore, the spacing would increase steadily. If convection due to surface tension gradients were suddenly ''switched on'' starting with this initial hypothetical temperature distribution, the large temperature gradients near the keyhole would cause a large radial acceleration of the melt near the keyhole surface towards the outer boundary. This radial convective heat transfer reduces the temperature gradients near the keyhole, thus increasing the isotherm spacing. Convection transports hot fluid towards the outer weld-pool boundary which causes the gradient to increase in that region compared to the initial situation. Consequently, the mechanism of heat convection due to Marangoni flow ($\partial\gamma/\partial T<0$) acting in the same direction as the heat conduction has a smoothing effect on the temperature gradient throughout the surface flow field.

The now larger temperature gradients near the outer boundary cause a larger heat flux into the solid. Therefore, the weld pool is extended in the upper region compared to a flow field without Marangoni convection.

Slightly below the surface, there must be a flow that moves back towards the keyhole for continuity reasons under the assumption of a fixed weld-pool surface. In this fluid layer, convection acts in the opposite direction of heat conduction. Thus, the radial heat transport from the keyhole towards the outer boundaries is reduced there. On the right in Fig. 7, one can see very closely spaced isotherms just behind the keyhole only slightly below the surface. This tight isotherm spacing behind the keyhole leads to a locally large heat flux but the heat is convected to the top and not towards the outer boundary in this fluid layer (see the arrow in the left figure). Due to this radially reduced heat transfer, the solidification isotherm at the weld-pool boundary has a step-like shape there. The weld pool underneath the recirculating fluid layer does not show much convection and, therefore, the isotherm distribution looks similar to one of mere heat conduction.

It should be noted that most of the significant flow activity is limited to the top of the pool (here, about 1 mm). If additional material like filler wire without surface-active agents is applied, it is not expected to mix with the lower part of the weld pool.

2. Positive $\partial\gamma/\partial T$

The case of a positive $\partial\gamma/\partial T$ resembles the coefficient of surface tension of a steel with a considerable amount of one or more surfactants like sulfur. In Fig. 8 the surface of the flow field is presented in the top view. The isotherms are in the upper half of Fig. 8. Many of them are concentrated very closely spaced just behind the keyhole. This results in very large surface-tension gradients behind the keyhole which cause high velocities in that region. In the lower half of Fig. 8—where the velocity field is presented—the velocity vectors point towards the keyhole because the coefficient of surface tension increases with the temperature. The velocity is considerably higher than that in the simulation with negative $\partial\gamma/\partial T$ because of much larger surface tension gradients. The melt at the surface reaches maximum speeds of 156 m/min which is 78 times the welding speed.

The evolution of such large temperature gradients becomes clearer by looking at Fig. 9. At the surface, the melt is driven towards the keyhole. There is a pressure buildup at the rear of the keyhole which causes the melt to move downward along the keyhole wall where it heats up to vaporization temperature. After 25%–50% of the keyhole depth, this hot flow separates from the keyhole wall and moves towards the rear of the weld pool. In this lower region of the pool, the isotherm spacing is almost constant, and the isotherms are clearly bent to the right due to the same direction of the convective and conductive heat transport. This is also true for the solidification isotherm which inhibits the step-like

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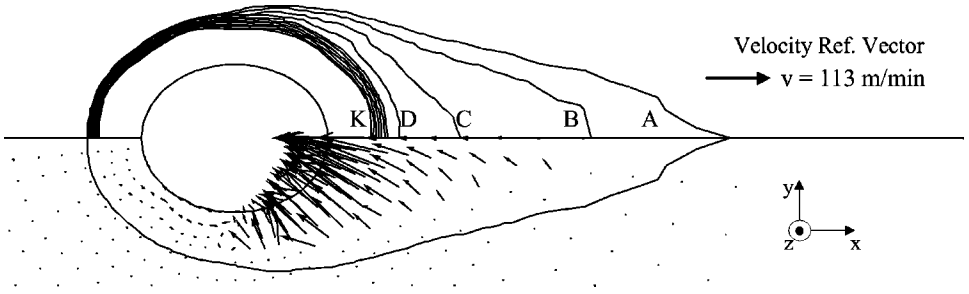


FIG. 8. Temperature distribution (represented by isotherms) and flow field at the surface in case of a positive $\partial\gamma/\partial T$. Temperatures of isotherms: $T_A = 1825$ K, $T_B = 1900$ K, $T_C = 2000$ K, $T_D = 2100$ K, $T_E = 2200$ K, $T_F = 2300$ K, $T_G = 2400$ K, $T_H = 2500$ K, $T_I = 2600$ K, $T_J = 2700$ K, and $T_K = 2770$ K (process and material parameters according to Table I).

shape at the rear of the weld pool in contrast to the previous simulation.

Due to continuity considerations, the flow has to move to the surface again, eventually. Once it reaches the surface, its temperature has decreased because of heat conduction. At the surface, convection acts in the opposite direction of heat conduction. Thus, radial heat transport is reduced which results in a decrease of the temperature gradients close to the solidification line and an increase of the temperature gradients close to the keyhole. This mechanism is possibly limited by friction and heat flux.

Just behind the keyhole, there is a temperature difference over one grid cell of 570 K which represents a distance of about 0.1 mm. It is well possible that this temperature gradient could increase even more with a finer mesh. With the temperature gradient, also the maximum velocity would increase in this area.

This simulation shows a much more deep reaching and shorter vortex than the simulation with negative $\partial\gamma/\partial T$. If additional material like filler wire were used, it should be much more uniformly distributed throughout the seam than in the first simulation with the rather shallow vortex.

In Sec. III C, an explanation will be given as to why the vortices at positive and negative $\partial\gamma/\partial T$ are so different in their shape and penetration depth.

C. Observations and interpretation of results

In order to explain the vortex structure, one needs to acknowledge the following three basic thoughts first:

(1) At low and medium welding speeds, the surface flow is mainly accelerated by the Marangoni effect. Thus, the surface flow always moves towards the isotherm where the coefficient of surface tension is highest. Other effects such as the flow around the keyhole are relatively unimportant in this range of welding speeds (has been tested up to 6 m/min). At high translation speeds, the flow around the keyhole may gain importance, though.

(2) In addition to the sign of $\partial\gamma/\partial T$, the velocity field is strongly dependent on the geometric constraints such as the weld-pool boundary and the keyhole since one or more velocity components are prescribed there [keyhole: $v_\perp = 0$; melting/solidification isotherm: $\mathbf{v} = (u_0,0,0)$]. Thus, all accelerations that lead to velocities that are beyond the required values for these boundary conditions need to be compensated by decelerations. With "thought (1)" in mind, it becomes clear that the pressure-based terms decelerate the surface flow and become important in front of obstacles. Their magnitude is, therefore, directly dependent on the previous acceleration due to the Marangoni effect.

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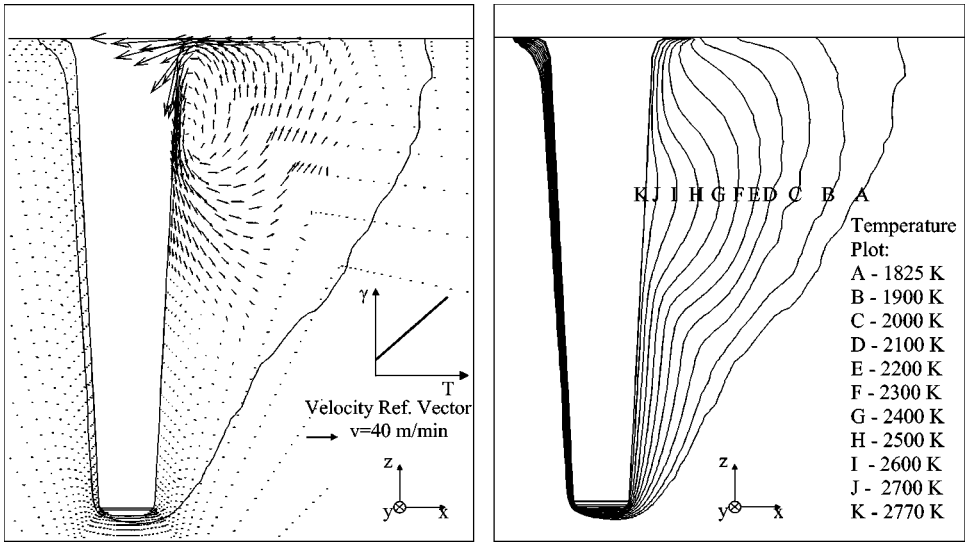


FIG. 9. Velocity vector plot and solidus isotherm in longitudinal section (left) and corresponding isotherm plot (right) in the case of positive $\partial\gamma/\partial T$ (corresponding to Fig. 8).

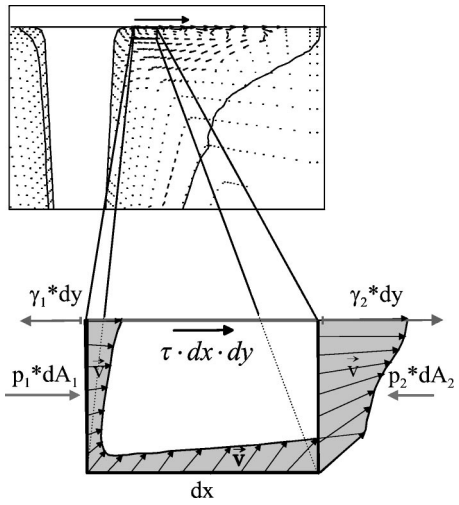


FIG. 10. Marangoni effect and pressure gradient accelerating the surface flow.

(3) At the surface, changes of the horizontal velocity lead to vertical velocity components underneath the surface. To investigate the reason for the vertical velocities, a small surface-volume element in the region of acceleration has been isolated in Fig. 10. If the outflow velocity on the right is larger than the inflow velocity on the left, there must be a volume compensation from below or from the sides perpendicular to the flow direction. In other words: continuity can only be guaranteed if the difference in horizontal mass flux is compensated by incoming flux from underneath. The difference between inflow and outflow velocity in Fig. 10 stems from different accelerations of neighboring fluid elements, which are in turn caused by the gradient of the sum of surface-tension forces and pressure-based forces across the related fluid elements.

With the three thoughts in mind, the problem can now be explained: The fluid elements next to obstacles have the prescribed velocity component on the side of the obstacle. If there are accelerating or decelerating forces next to obstacles, there have to be large vertical velocities nearby [see ''thought (3)'']. At this point, only vertical velocities have been explained but not the vortices. Considering ''thought 2'' should help solve this problem. If all accelerations have to be compensated by decelerations as mentioned in thought 2, then all the vertical mass transfer to the surface that is not required to meet the boundary conditions has to be compensated by a flux away from the surface at some other location in the weld pool, which forms a vortex. In order to find the location and shape of the vortices, one only needs to know the locations of acceleration and deceleration which are shown in Fig. 11 for the related problems.

For the case of negative $\partial\gamma/\partial T$ (Fig. 11, on the left), the acceleration is near the keyhole which leads to an upward pointing vertical flow. The region of deceleration is close to the weld pool's rear which leads in a flow vector pointing downward, creating a rather long vortex. Since the acceleration as well as the deceleration extends over a relatively large

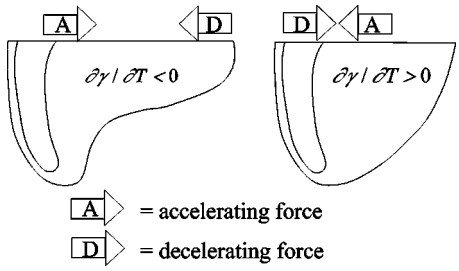


FIG. 11. Zones of acceleration and deceleration in weld pool at negative (left) and positive $\partial\gamma/\partial T$ (right).

distance the vertical velocity at the pool's rear is not very high. This situation and the step-like geometry lead to a rather shallow vortex.

The case of positive $\partial\gamma/\partial T$ is depicted in Fig. 11 on the right. Due to the large temperature gradient near the keyhole, the acceleration towards the keyhole is very large, but just next to this location the keyhole acts as an obstacle which leads to a pressure buildup and to deceleration. Since both acceleration and deceleration are adjacent to each other and are very intense, a short and deep reaching vortex is formed.

If the keyhole wall were not kept fixed in this model, the flow might overcome the forces that keep the keyhole open. Thus, the hot melt might run into the high-intensity laser beam where it would vaporize instantly. This may cause blow outs and splattering which is commonly observed in welding steels with a high content of sulfur.

In Fig. 12, the simulated cut sections close behind the keyhole ($1/2 \cdot a$ behind the keyhole according to Table I) are shown where the weld pools approximately have their largest cross section. The flow fields in Fig. 12 can be explained with the same considerations but the results will not be discussed in more detail.

D. Shortcomings of simulations with fixed surfaces

One should keep in mind that both the keyhole surface and the weld-pool surface are fixed in these simulations. If the keyhole wall were not kept fixed, the keyhole would have a different shape for the following reasons:

- It has been mentioned—without quantifying it so far—that the coefficient of surface tension partly depends on the sulfur concentration within the steel. This will have an impact on the pressure equilibrium at the keyhole wall.
- Surfactants lower the binding forces between molecules or atoms leading to a lower vaporization temperature and a lower vaporization enthalpy. If a constant heat flux is absorbed at the keyhole surface the recoil pressure due to vaporization should increase to some extent in the case of surfactants in the melt.
- In the case of a positive $\partial\gamma/\partial T$ the fluid convects heat along the keyhole wall towards the bottom of the pool. Therefore, less power is needed to reach vaporization temperature at such ''hot spots'' than at negative $\partial\gamma/\partial T$ where the heat convects to the surface. Thus, vaporization is increased at the back and at the bottom of the keyhole which

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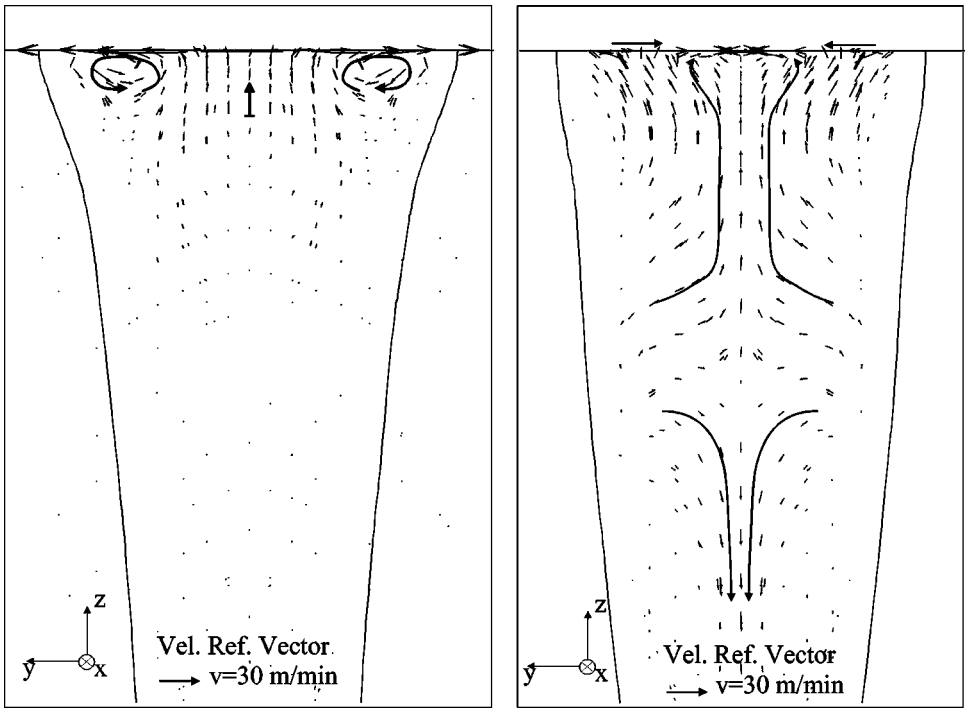


FIG. 12. Velocity-vector plot and solidus isotherm in enlarged cut section at $1/2 \cdot a$ behind the keyhole according to Table I for negative $\partial\gamma/\partial T$ (left) and the corresponding cut section for positive $\partial\gamma/\partial T$ (right).

leads to an increase in penetration depth and might account for an increase of process pores at a positive slope $\partial\gamma/\partial T$ compared to a negative one.

- As mentioned in Sec. III B 2, the fast flow which occurs at positive $\partial\gamma/\partial T$ might run into the laser beam or at least deform the keyhole. This could lead to blow outs or splattering and a decrease in keyhole diameter. Moreover, this keyhole deformation could affect the penetration depth.

E. Dependence of $\partial\gamma/\partial T$ on the sulfur content

After the discussion of the above-mentioned simulation results, the following questions arise:

(1) Which sulfur content is needed for steel to invert the slope of the coefficient of surface tension?
(2) Which steels have a positive or a negative sign of $\partial\gamma/\partial T$?

An answer to question (1) can be found in the literature, e.g., in Ref. 13. The results are displayed in Fig. 13. Figure 13 shows that for steel without sulfur (or pure iron) the coefficient of surface tension decreases linearly over the temperature and intersects the x axis at the critical temperature. This case corresponds to the negative $\partial\gamma/\partial T$ used in the simulation in Sec. III B 1. At 500 ppm the slope is almost inverted throughout most of the temperature range between melting and vaporization. This case resembles the positive $\partial\gamma/\partial T$ used in the simulation in Sec. III B 2.

In order to answer the second question, the tolerated ranges for the sulfur content c_s (the index s indicates sulfur) for some typical material numbers are listed in the following:

- 16MnCr5: $0 \leqslant c_s \leqslant 350$ ppm,
- St52: $0 \leqslant c_s \leqslant 500$ ppm, and
- A1320: $0 \leqslant c_s \leqslant 400$ ppm.

If one compares these very typical fields of tolerances for certain steels with the dependence of γ on the temperature and on the concentration of sulfur according to Fig. 13, it becomes obvious that the tolerances for the chemical additives are too crude for many steels in the context of reliably predicting the welding geometry. In the case of the steel St52 the curve of the temperature dependence of γ must be somewhere between the two curves of no sulfur and 500 ppm sulfur. Thus, the second question cannot be answered by just referring to a certain material number.

IV. CONCLUSION

The simulations demonstrate that the Marangoni effect is very important in laser deep penetration welding of steel, too. They show large differences in flow behavior merely by

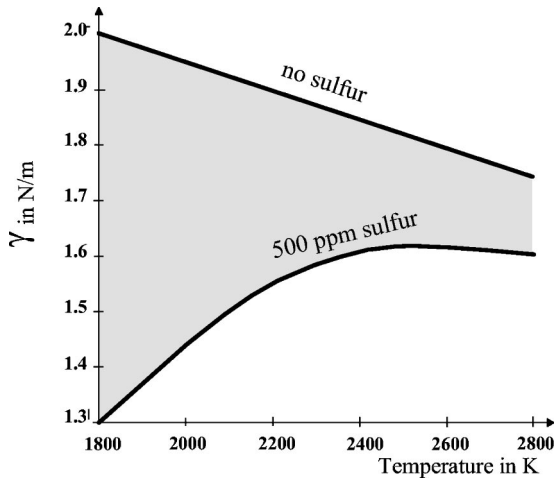


FIG. 13. Temperature dependence of the coefficient of surface tension γ according to Vitol and Orlova (Ref. 13).

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inverting the temperature dependence of γ. This leads to a different convective heat transfer and, therefore, the weld-pool shape can change dramatically. The shape of the weld pool determines the cooling rates in the solidus–liquidus zone and influences the crystal growth there.

Unfortunately, the tolerances in sulfur content of typical steel standards are so crude that both a positive and a negative surface-tension gradient can occur.

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